The AI-GPR Index: Measuring Geopolitical Risk using Artificial Intelligence

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Measuring geopolitical risk: A new tool

- Economists have long tried to quantify hard-to-measure phenomena by counting specific word combinations newspapers — cheap and transparent. E.g. BBD's EPU Index, or GPR Index of Caldara-Iacoviello
- Limitation of these approaches: dictionary-based approach cannot easily tell whether "war" in an article means active conflict or retrospective on WWII; cannot say who is threatening whom; cannot tell whether crisis threatens oil supplies.
- An LLM can read and score each article, as a research assistant would, at a scale no team of assistants could match.

What this opens up

Better measurement of the same concept (less noise, fewer false positives); a sharper version of Caldara-Iacoviello's threats-vs-acts split; and dimensions word-counting cannot reach: oil-specific risk, directed bilateral stress

The AI-GPR index

What we do

We build a daily measure of geopolitical risk from 4.6 million newspaper articles spanning 1960 to present. Each article is read and scored by an AI model (GPT-4o mini).

We extend the same approach with additional classification layers: one refines an existing distinction, two reach dimensions keyword-based methods cannot produce.

1. The AI-GPR Index

- Uses definition of geopolitical risk from Caldara and Iacoviello (2022)
- LLM scores every article 0–1 for geopolitical risk intensity
- Available daily from 1960 onwards

2. AI-GPR Measures

- **Threats and Acts**: is the risk anticipated (threat) or realized (act)?
- **Oil GPR**: does this article discuss a geopolitical oil supply disruption, and if so where?
- **Bilateral GPR**: who is acting on whom: initiator, respondent, spillover countries

Step 1: Building the index

Collecting Articles

Collect every article that might discuss geopolitical risk

Article query:

```
airstrike* OR alliance OR annex* OR attack* OR
blockade* OR bomb* OR cease-fire OR combat OR
conflict* OR coup OR crisis OR diplomat* OR embargo
OR enem* OR hostage* OR hostil* OR invade* OR
invasion* OR militia OR military OR missile*
OR nuclear OR refugee* OR riot OR sanction* OR
sovereign OR terror* OR treaty OR troop* OR truce
OR unrest OR violence OR war OR weapon* ...
```

Numbers:

- 4.6 million pass the article query filter
- 8.5 million total articles (NYT, WaPo, Chicago Tribune, 1960–2025)
- Filter is deliberately inclusive; false positives handled in Step 2

Step 2: Full classification prompt (simplified)

Pass each article to the LLM and score its geopolitical risk content

```
You will be given [the first 2,000 characters of] a news article.

Classify the article's assessment of geopolitical risk based only on what the article states or
strongly implies.

Geopolitical risk is defined as the threat, realization, and escalation of adverse events associated
with:  wars (...); escalation of existing wars (...); major terrorist attacks (...); and tensions
among states and political actors (...).

Assign a geopolitical risk score from 0.0 to 1.0 based on the following scale:

0.0--0.2:  No mention of geopolitical risks.
0.2--0.4:  Mentions of minor tensions, diplomatic disputes, or isolated incidents....
0.4--0.6:  Discussion of significant tensions...
0.6--0.8:  Substantial discussion of major war initiation/escalation risks....
0.8--1.0:  Extensive coverage of imminent or new war, major war escalation, severe terrorism threats,
or critical threat to international stability.

Movies, books, anniversaries of old events, and obituaries should receive a score of 0.0 ...
Purely domestic political events (elections, protests, internal policy debates) should score 0.0 ...
```

The scoring scale in practice: Results

Score: 0.2 (minor tensions)

Kim Jong Un takes his daughter to China — and fuels succession chatter

Kim Ju Ae, who is about 12, has accompanied her father on a high-profile visit to Beijing, making it more likely that she is North Korea's heir apparent.

September 3, 2025

Score: 1.0 (imminent or active war / severe threat)

The Legality, or Illegality, of Killing a Foreign Leader, Explained

The United States and Israel started a war with Iran by killing its supreme leader, Ayatollah Ali Khamenei.

Listen · 8:48 min

Share full article



Calculating the AI-GPR index

Daily index formula

$$\text{AI-GPR}_t = \sum_{i=1}^{N_t} S_{it} \times \frac{1}{A_t} \times \frac{1}{\bar{S}}$$

- S_{it} = GPR score assigned by LLM to article i on date t
- N_t = number of scored articles on date t (those passing the filter)
- A_t = Total articles published on date t (denominator, not just filtered)
- $\bar{S}$ = normalization constant

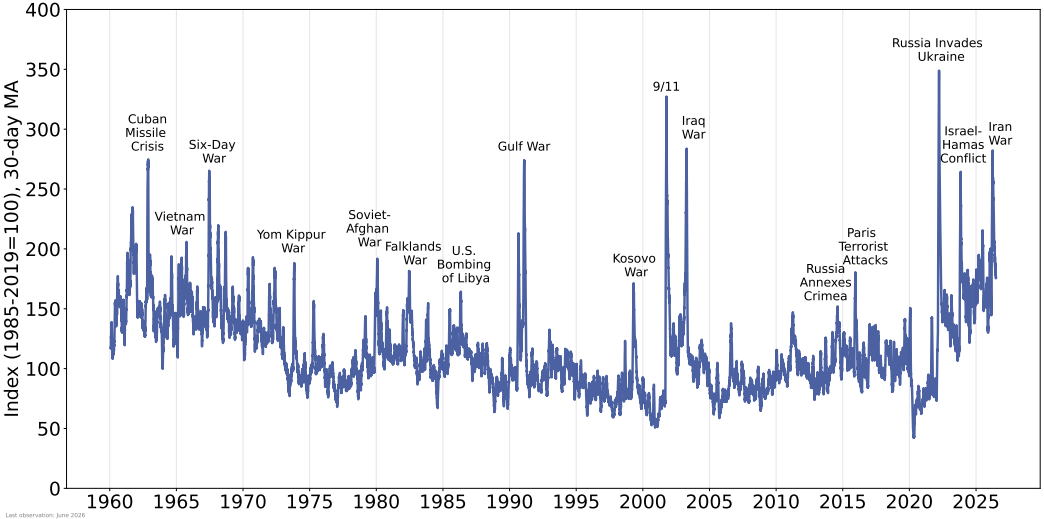
Result of Steps 1+2:

- 4.6M articles filtered
- 1.2M (26%) receive score > 0
- 14% of total articles are GPR-relevant

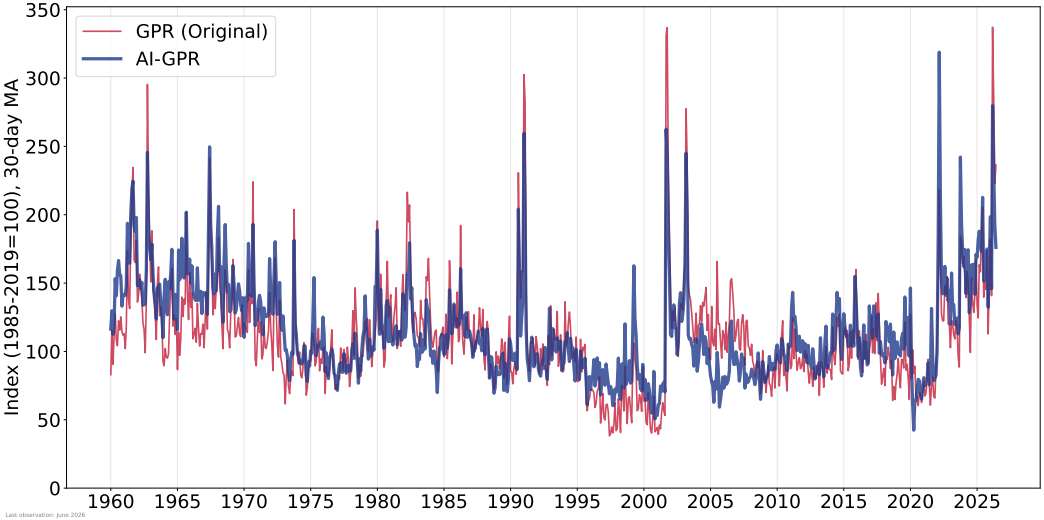
Key properties:

- Daily frequency, 1960–2026
- Continuous intensity (not binary)
- Correlation 0.69 with original GPR; higher persistence (autocorr. 0.73 vs. 0.62)

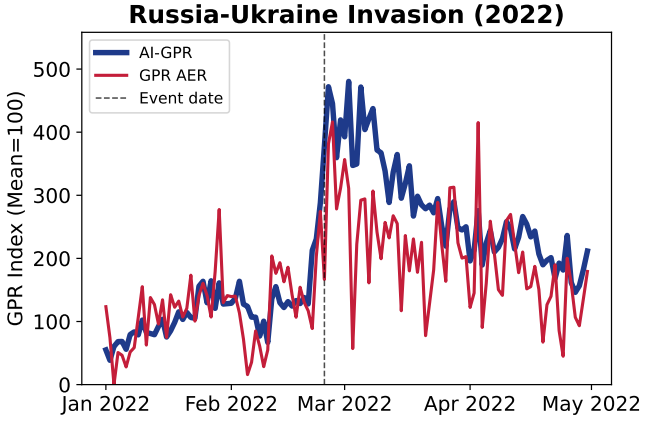
The AI-GPR index, 1960–2026



AI-GPR vs. original GPR: Monthly comparison (1960-2026)

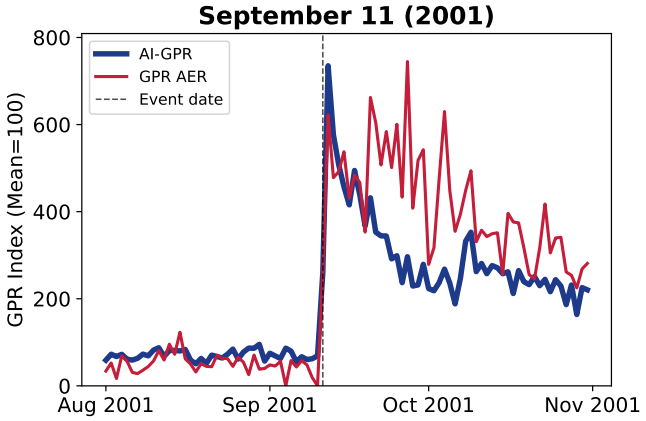


How accurate is the AI-GPR? Russia's invasion of Ukraine (2022)



Daily AI-GPR (blue) and keyword-based GPR (red) around Russia's invasion of Ukraine, adjusted for day-of-the-week effects. The AI-GPR spikes sharply on invasion day with less day-to-day noise.

How accurate is the AI-GPR? September 11 attacks (2001)



Daily AI-GPR (blue) and keyword-based GPR (red) around the September 11 attacks, adjusted for day-of-the-week effects. Both spike on the day of the attacks; the AI-GPR then declines steadily, while the keyword-based GPR stays noisier.

Measurement validation

Concern	Test	Finding
Stochasticity	Re-run 3x on 1,050 articles	Correlation 0.99 across runs
Text truncation	2,000-char vs. full text	Correlation 0.93; 87% identical scores
Bias vs. humans	Human audit: 2,447 articles, stratified across the whole corpus (2,101 double-coded)	Correlation with human consensus: 0.85 Correlation between humans: 0.78 False-discovery rate 13% (keyword-based: 18%) Miss rate 24% (keyword-based: 82%)
Model sensitivity	GPT-4o, GPT-5, Llama 3.1	Cross-model correlation 0.84–0.94
Hindsight bias	Date-anchored prompt: model told it knows nothing after the article's date	Correlation with baseline 0.95
Keyword filter misses	LLM on 10,000 excluded articles	0.6% receive positive score (keyword GPR: 2.6%)

Audit rates weighted by corpus share of each stratum. Miss rate: human-positive articles not flagged. False-discovery rate: flagged articles a human scores zero.

Three ‘extra’ measures

1. Threats vs. Acts

CI2022’s split, sharpened by AI

→ stock returns

2. Oil disruptions

Geopolitical oil supply disruptions, by region

→ proxy SVAR

3. Bilateral risk

Who acts on whom, directed by country pair

→ bilateral trade

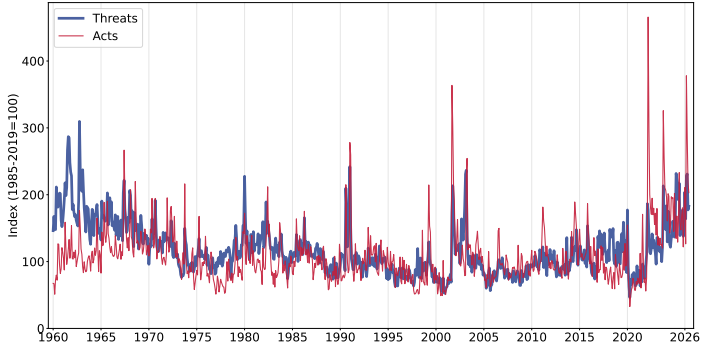
Each measure reuses the same pipeline: a second LLM classification applied on top of the headline AI-GPR score.

Measure 1/3: Threats vs. acts

A second classification layer

Caldara-Iacoviello (2022) already split GPR into threats and acts by keyword

We rebuild the same distinction with an LLM. Each article with a positive GPR score is re-read and labeled a **threat** (anticipated risk), an **act** (event occurred or underway), or **both** (∼15%).



Application 1: Geopolitical risk and stock returns

	Weekly excess return, %	
	(1) OLS	(2) Decomp.
ΔGPR_t	-0.136^{**}	
	(0.057)	
Persistent $\widehat{\Delta \text{GPR}_t}$		-0.271^{**}
		(0.134)
Shock $\hat{u}_t$		-0.119^{*}
		(0.062)
Obs.	3,452	3,448
Sample	1960–2026	

Weekly excess returns (%). Pairs-bootstrap SE in col. (2).

GPR and stock returns (col.1)

- Analyze relationship between changes in GPR and weekly stock returns
- OLS coefficient is negative and significant
- Same exercise with keyword-based GPR: OLS coefficient imprecisely estimated $\Rightarrow$ AI-GPR is more precisely estimated

Decomposing stock returns (col.2)

- Decompose ΔGPR into:
  - persistent: AR(4) fitted values
  - shock (residuals) component
- Persistent component coefficient (-0.27) is twice aggregate OLS (-0.14):
- Markets price persistent component of risk, not just weekly surprises

Threats vs. acts: Priced differently

	Weekly excess return, %	
	(1) OLS	(2) Decomp.
$\Delta\text{GPR}_{threats}$	-0.081^{**}	
	(0.037)	
ΔGPR_{acts}	-0.084	
	(0.061)	
Persistent, threats		-0.220^{**}
		(0.103)
Persistent, acts		-0.063
		(0.131)
Shock, threats		-0.052
		(0.044)
Shock, acts		-0.090^{**}
		(0.042)

Weekly excess returns (%), 1960–2026.

In levels (col. 1)

- Threats: precisely estimated negative effect
- Acts: similar size, not significant

Decomposed (col. 2)

- Threats matter through **predictable** component: markets re-price as a threat persists
- Acts matter through their **surprise** component

Takeaway

Whether risk is anticipated or unexpected matters more than its level. Splitting threats from acts makes this visible.

Measure 2/3: Geopolitical oil supply disruptions

Two-stage LLM classification

Stage 1: every article receives a GPR score 0–1 (AI-GPR)
Stage 2: articles with GPR > 0.5 that contain oil keywords are re-classified: does this article discuss an *oil supply disruption caused by a geopolitical event*? If yes, which region?

Coverage (of articles with GPR > 0.5)

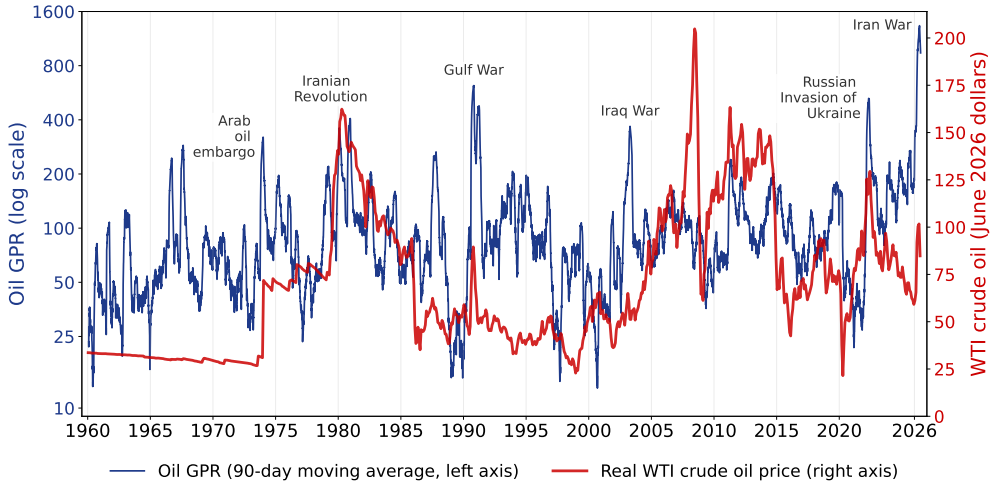
- 12.6% contain oil keywords
- 9.3% classified as oil disruption

Regional breakdown

- Middle East: 63%
- Russia: 14% (surges after 2014)
- North Africa: 11%
- Venezuela, W. Africa, SE Asia: 5–6% each

Oil $\text{GPR}_t = \sum_{i \in \text{disruptions}} S_{it} \;/\; A_t$ (weighted by GPR intensity, normalized by total articles)

Oil GPR index and oil prices



Oil GPR index (left axis, 90-day MA) and real WTI oil price (monthly, right axis). Sample ends June 30, 2026

Application 2: A proxy SVAR with geopolitical oil market shocks

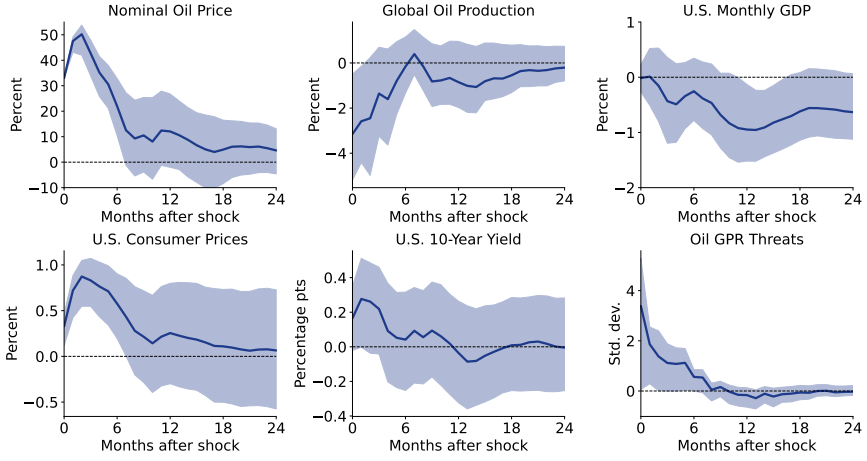
External instrument

- News series: Oil GPR *threats* (oil-disruption articles classified as threats)
- Event days: index > 1 s.d. above its mean
- Instrument: WTI log-price change from the last close before to the first close after each event day, summed by month
- These days have a clear geopolitical trigger, not an OPEC decision, a demand swing, or a weather event

VAR setup

Proxy SVAR with: nominal U.S. crude oil price, global oil production, U.S. monthly GDP, U.S. CPI, 10-year Treasury yield, and the Oil GPR threats index. 12 lags, monthly, January 1985 to June 2026. First-stage $F = 38$.

Proxy SVAR: Impulse responses to a geopolitical oil market shock



Impulse responses to a geopolitical oil market shock, normalized to a 33 log point oil price rise on impact. Shaded bands: 80% moving block bootstrap CI (Jentsch–Lunsford), 6-month blocks. 12 lags. Sample: January 1985 to June 2026.

Measure 3/3: Who acts on whom? Bilateral country geopolitical risk

A third classification layer

Every article with a positive GPR score is re-read to identify up to five country-level actors and assign each a role: **initiator** (launched the action), **respondent** (direct target), or **spillover** (affected, not directly involved).

Three roles

- **Initiator**: launched or triggered the action
- **Respondent**: direct object of the action
- **Spillover**: affected but not directly involved

What it produces

- Country-level GPR for 200 countries
- Directed bilateral GPR for the top 1,200 country pairs

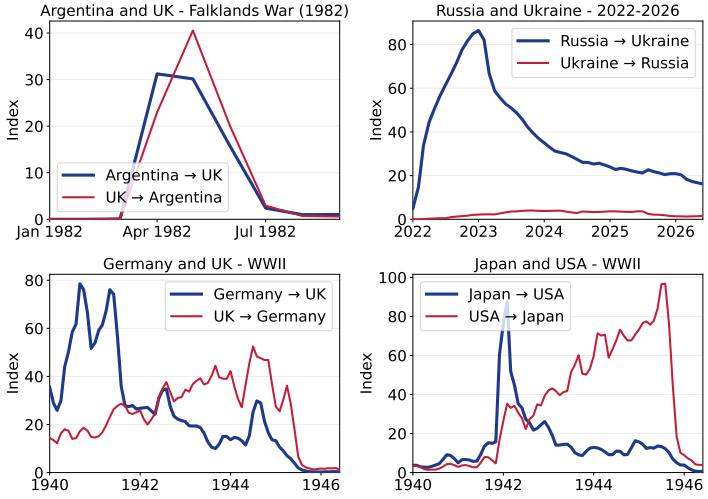
Bilateral geopolitical risk: A directed measure

What it Measures

For each country pair (i, j): share of articles where i is *initiator* and j is *respondent*.

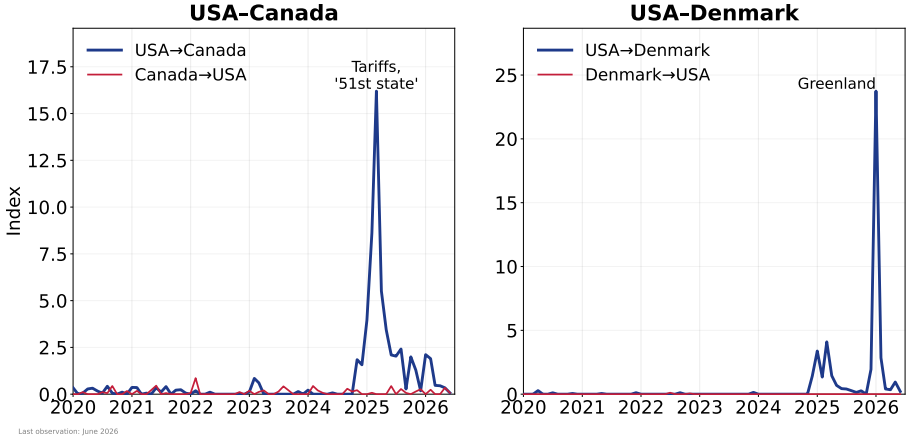
Directed:
Russia acting on Ukraine $\neq$ Ukraine acting on Russia.

- Extends CI2022 country indexes to country pairs
- Constructed for 1,200 country pairs
- (in progress) Going back all the way to 1900



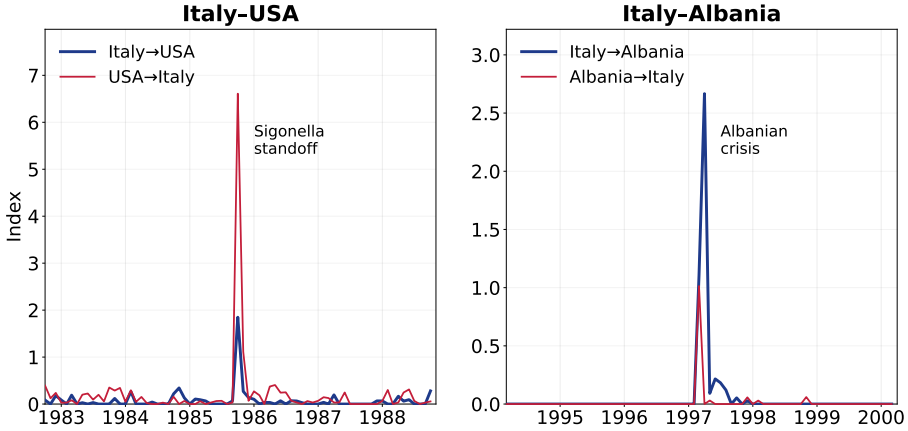
Selected pairs by bilateral GPR

Bilateral GPR since 2020: The U.S. and two allies



Monthly bilateral AI-GPR, not smoothed. Thick blue: United States as initiator. Thin red: reverse direction.

Bilateral GPR: Italy's two largest spikes



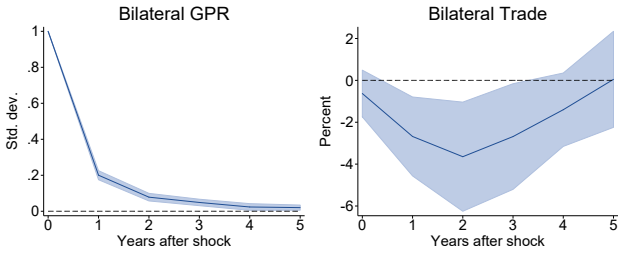
Monthly bilateral AI-GPR, not smoothed. Thick blue: Italy as initiator. Thin red: reverse direction. Pairs with the largest one-month increases involving Italy.

Application 3. Does bilateral GPR depress bilateral trade?

Local Projection

$$y_{od,t+h} - y_{od,t-1} = \gamma_h Z_{od,t} + ... + \varepsilon_{od,t+h}$$

- $y_{od,t} = 100 \times$ log two-way trade
- $Z_{od,t}$: bilateral AI-GPR, standardized
- Pair FE, and year FE for each country in the pair.
- Controls: three lags of GPR, three lags of y, trade agreement.



Response to a one-standard-deviation rise in bilateral GPR, relative to year $t-1$. Shaded areas are 90% confidence intervals, standard errors clustered by each country in the pair and by year. Comtrade, 1962–2020.

Argentina–U.K. trade around the Falklands War



Source: Comtrade via CEPII Gravity V202211, deflated to constant 2012 USD.

Conclusions

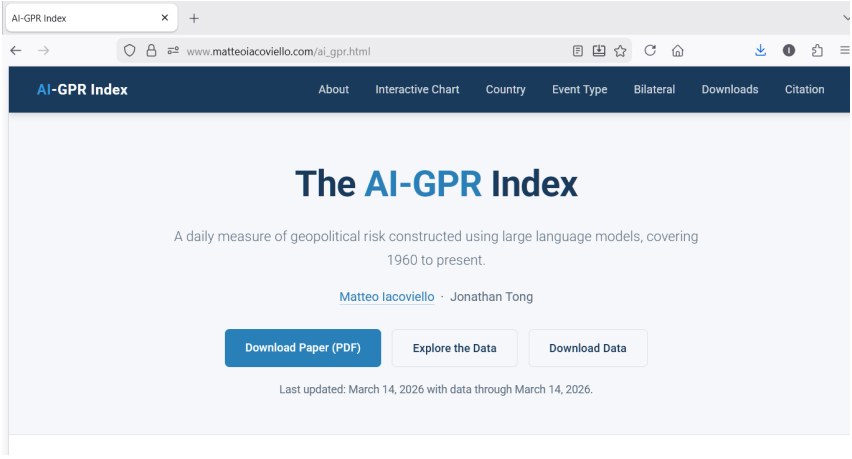
The AI-GPR Index

We construct an LLM-based measure of geopolitical risk that is more precise than keyword methods and measures things keyword methods cannot

Three Applications

- **Stock returns**: AI-GPR gives a more precise negative effect. Threats are priced when anticipated, acts when unanticipated
- **Oil shocks**: two-stage LLM identifies 65 years of geopolitical oil disruptions; geopolitical oil market shocks identified on days of oil-disruption threats raise inflation and lower GDP
- **Bilateral trade**: directed country-pair indexes (who acts on whom); averaged over both directions, bilateral GPR predicts trade declines

All data available at matteoiacoviello.com/ai_gpr.html



Daily and monthly indexes, country decompositions, bilateral pairs, event types — all downloadable. Interactive charts with smoothing controls.